faster and faster, there is no sign at all of any swirl. Some atoms circle the hole clockwise as they fall; others circle it counterclockwise and occasionally collide with clockwise-circling atoms; but on average the atoms' fall is directly inward (directly downward) with no swirl. Your conclusion: This 10-solar-mass black hole is hardly spinning at all; its angular momentum is close to zero.

Knowing the mass and angular momentum of the hole and knowing that its electrical charge must be negligibly small, you can now compute, using general relativistic formulas, all of the properties that the hole should have: the strength of its gravitational pull, its corresponding power to deflect starlight, and of greatest interest, the shape and size of its horizon.

If the hole were spinning, its horizon would have well-delineated north and south poles, the poles about which it spins and about which infalling atoms swirl. It would have a well-delineated equator halfway between the poles, and the centrifugal force of the horizon's spin would make its equator bulge out,[14] just as the equator of the spinning Earth bulges a bit. But Hades spins hardly at all, and thus must have hardly any equatorial bulge. Its horizon must be forced by the laws of gravity into an almost precisely spherical shape. That is just how it looks through your telescope.

As for size, the laws of physics, as described by general relativity, insist that the more massive the hole is, the larger must be its horizon. The horizon's circumference, in fact, must be 18.5 kilometers multiplied by the mass of the hole in units of the Sun's mass.[15] Since your orbital measurements have told you that the hole weighs ten times as much as the Sun, its horizon circumference must be 185 kilometers—about the same as Los Angeles. With your telescopes you carefully measure the circumference: 185 kilometers; perfect agreement with the general relativistic formula.

This horizon circumference is minuscule compared to your starship's 1 -million-kilometer orbit; and squeezed into that tiny circumference is a mass ten times larger than that of the Sun! If the hole were a solid body squeezed into such a small circumference, its average density would be 200 million (2×10^8) tons per cubic centimeter—2×10^{14} times more dense than water; see Box P.1. But the hole is not a solid body. General relativity insists that the 10 solar masses of stellar matter, which created the hole by imploding long ago, are now concentrated at the hole's very center—